

Solution to MP 363 Exam '14-'15

(1)

P. 1

We see that the state given may be rewritten as $|\psi\rangle = \sum_{n=1}^{\infty} \alpha_n |\phi_n\rangle$, where $\alpha_n = A/n$. This will prove to be useful for (a) & (b)...

(a) Recall that if we write a state in terms of orthonormalised states,

like we have above, the condition $\|\psi\| = 1$ (i.e. $\langle\psi|\psi\rangle = 1$) is equivalent to $\sum_{n=1}^{\infty} |\alpha_n|^2 = 1$. Furthermore we have here, $|\alpha_n|^2 = \frac{A^2}{n^2}$

(since A is real and positive, $|A| = A$). Thus, normalisation is equivalent to $\sum_{n=1}^{\infty} \frac{A^2}{n^2} = 1$. But $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ is given, so that we have $\frac{\pi^2 A^2}{6} = 1$.

Since A is real and positive, $A = \sqrt{A^2}$, so

$$A = \sqrt{A^2} = \sqrt{6/\pi^2} = \frac{\sqrt{6}}{\pi}$$

(20pts)

(b) We may also use these coefficients to compute the expectation value of H : recall that if $|\phi_n\rangle$ has energy E_n , and a state $|\psi\rangle$ is written as $\sum_{n=1}^{\infty} \alpha_n |\phi_n\rangle$, then the expectation value of H is

$$\langle H \rangle = \sum_{n=1}^{\infty} |\alpha_n|^2 E_n$$

We have α_n , and the E_n are given by E_1/n^2 , so we find

$$\begin{aligned} \langle H \rangle &= \sum_{n=1}^{\infty} |\alpha_n|^2 E_n = \sum_{n=1}^{\infty} \frac{A^2}{n^2} \cdot \frac{E_1}{n^2} = A^2 E_1 \sum_{n=1}^{\infty} \frac{1}{n^4} \\ &= \frac{\pi^4 A^2 E_1}{90} = \frac{\pi^2 E_1}{15} \end{aligned}$$

since $A^2 = 6/\pi^2$. Putting in $E_1 = -13.6 \text{ eV}$ and doing the number-crunching gives

$$\langle H \rangle = -8.95 \text{ eV.}$$

(15pts)

(c) We know that a measurement of the energy can return only one of the energy eigenvalues E_n ; the question is, which one of these is -0.85 eV ? This is straightforward: we look for the n such that $E_n = \frac{E_1}{n^2} = -0.85 \text{ eV}$, i.e.

$$n = \sqrt{\frac{E_1}{-0.85 \text{ eV}}} = \sqrt{\frac{-13.6 \text{ eV}}{-0.85 \text{ eV}}} = \sqrt{16} = 4$$

so the value we get is E_4 . But the postulates of QM tell us that immediately after we get a value for the

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energy, the state becomes the eigenstate with that energy.

Thus, getting an energy of E_4 rec_g that the state "collapses" to $|\phi_4\rangle$. (15 pts)

P. 2

The general defn of the adjoint goes as follows: If A is an operator acting on a Hilbert space $\mathcal{H}$, the $A^\dagger$ is that operator such that $\langle A^\dagger \psi | \zeta \rangle = \langle \psi | A \zeta \rangle$ for all $\psi, \zeta \in \mathcal{H}$. However, in this problem, we deal only with $n \times n$ complex matrices. ...

(a) If M is an $n \times n$ complex matrix, the adjoint operation consists of taking the transpose of M and then complex-conjugating all the entries in M , i.e.

$$\boxed{M^\dagger = (M^T)^* = (M^*)^T} \quad (4 \text{ pts})$$

(You can do the transposition and complex-conjugation in either order.)

(b) A self-adjoint operator is any operator such that it is its own adjoint, i.e. $A^\dagger = A$. In the case of matrices, that is equivalent to $(M^*)^T = M$, so we want to check this for the three matrices given:

$$\sigma_1^\dagger = (\sigma_1^T)^* = \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}^T \right]^* = \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right]^* = \begin{pmatrix} 0^* & 1^* \\ 1^* & 0^* \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \boxed{= \sigma_1} \quad (4 \text{ pts})$$

$$\sigma_2^\dagger = (\sigma_2^*)^T = \left[\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}^* \right]^T = \left[\begin{pmatrix} 0^* & (-i)^* \\ i^* & 0^* \end{pmatrix} \right]^T = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}^T$$

$$= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \boxed{= \sigma_2} \quad (4 \text{ pts})$$

$$\sigma_3^\dagger = (\sigma_3^T)^* = \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}^T \right]^* = \left[\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right]^* = \begin{pmatrix} 1^* & 0^* \\ 0^* & (-1)^* \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \boxed{= \sigma_3} \quad (4 \text{ pts})$$

so all three matrices are self-adjoint.

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(c) The inner product on the space is $\langle M|N \rangle = \frac{1}{2} \text{tr}(M^\dagger N)$; we set to show that $\langle \sigma_i | \sigma_j \rangle = \delta_{ij}$, as this is the statement of orthonormality of the vectors in (b).

Now, since $\langle M|N \rangle$ is an inner product, $\langle M|N \rangle^* = \langle N|M \rangle$, which implies if $\langle \sigma_i | \sigma_j \rangle = 0$, then $\langle \sigma_j | \sigma_i \rangle = 0$ as well, so we only need compute the real inner products $\langle \sigma_1 | \sigma_1 \rangle$, $\langle \sigma_1 | \sigma_2 \rangle$, $\langle \sigma_1 | \sigma_3 \rangle$, $\langle \sigma_2 | \sigma_2 \rangle$, $\langle \sigma_2 | \sigma_3 \rangle$ and $\langle \sigma_3 | \sigma_3 \rangle$ to check orthonormality. Also, in (b) we showed $\sigma_i^\dagger = \sigma_i$, so $\langle \sigma_i | \sigma_j \rangle = \frac{1}{2} \text{tr}(\sigma_i^\dagger \sigma_j) = \frac{1}{2} \text{tr}(\sigma_i \sigma_j)$.

Thus, we need to confirm that:

$$\frac{1}{2} \text{tr}(\sigma_1^2) = \frac{1}{2} \text{tr}(\sigma_2^2) = \frac{1}{2} \text{tr}(\sigma_3^2) = 1$$

and

$$\frac{1}{2} \text{tr}(\sigma_1 \sigma_2) = \frac{1}{2} \text{tr}(\sigma_1 \sigma_3) = \frac{1}{2} \text{tr}(\sigma_2 \sigma_3) = 0.$$

Let's do this:

$$\frac{1}{2} \text{tr}(\sigma_1^2) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = \frac{1}{2}(1+1) = 1 \quad [5 \text{pts}]$$

$$\frac{1}{2} \text{tr}(\sigma_2^2) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\right) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = \frac{1}{2}(1+1) = 1 \quad [5 \text{pts}]$$

$$\frac{1}{2} \text{tr}(\sigma_3^2) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = \frac{1}{2}(1+1) = 1 \quad [5 \text{pts}]$$

$$\frac{1}{2} \text{tr}(\sigma_1 \sigma_2) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\right) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}\right) = \frac{1}{2}(i-i) = 0 \quad [5 \text{pts}]$$

$$\frac{1}{2} \text{tr}(\sigma_1 \sigma_3) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\right) = \frac{1}{2}(0+0) = 0 \quad [5 \text{pts}]$$

$$\frac{1}{2} \text{tr}(\sigma_2 \sigma_3) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right) = \frac{1}{2} \text{tr}\left(\begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}\right) = \frac{1}{2}(0+0) = 0 \quad [5 \text{pts}]$$

so $\langle \sigma_i | \sigma_j \rangle = \delta_{ij}$ does indeed hold.

P.3

The given state $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle - \frac{i}{2}|1\rangle$ is written in terms of the energy eigenstates $|0\rangle, |1\rangle, |2\rangle, \dots$ as $|\psi\rangle = \sum_{n=0}^{\infty} \alpha_n |n\rangle$

with $\alpha_0 = \frac{\sqrt{3}}{2}$, $\alpha_1 = -\frac{i}{2}$ and all other coefficients are zero. (Note that $\sum_{n=0}^{\infty} |\alpha_n|^2 = |\alpha_0|^2 + |\alpha_1|^2 = \left|\frac{\sqrt{3}}{2}\right|^2 + \left|-\frac{i}{2}\right|^2 = \frac{3}{4} + \frac{1}{4} = 1$)

④

as it must.) These coefficients will be key in answering (a) and (b) ...

(a) Recall that if $|\psi\rangle = \sum_{n=0}^{\infty} \alpha_n |\phi_n\rangle$, the $|\alpha_n|^2$ is the probability for the value E_n will be obtained upon measuring the energy. For our case, $|\phi_n\rangle = |n\rangle$, $\alpha_0 = \frac{\sqrt{3}}{2}$, $\alpha_1 = \frac{1}{2}$ and $\alpha_n = 0$ for $n=2, 3, \dots$. Since $|\alpha_n|^2 = 0$ for $n=2, 3, \dots$, this means that we can never get $E_2, E_3, \dots$ as an energy.

Thus, the only possible values we can obtain are $E_0 = \frac{1}{2} \hbar \omega$ and $E_1 = \frac{3}{2} \hbar \omega$. The probabilities [4pts each]

of getting these are $|\alpha_0|^2$ and $|\alpha_1|^2$ respectively, so we see that the chance of getting $\frac{1}{2} \hbar \omega$ is $3/4$ (75%) and the chance of getting $\frac{3}{2} \hbar \omega$ is $1/4$ (25%). [4pts each]

(b) Like in problem 1, if we have the coefficients α_n , the the energy expectation value is $\langle H \rangle = \sum_n |\alpha_n|^2 E_n$. Here, only $n=0$ and $n=1$ have nonzero coefficients, so

$$\langle H \rangle = \sum_{n=0}^{\infty} |\alpha_n|^2 E_n = |\alpha_0|^2 E_0 + |\alpha_1|^2 E_1$$

$$= \left(\frac{3}{4}\right) \left(\frac{1}{2} \hbar \omega\right) + \left(\frac{1}{4}\right) \left(\frac{3}{2} \hbar \omega\right)$$

$$= \frac{3}{4} \hbar \omega$$

[4pts]

(c) This is rather similar to Problem 3 on Problem Set 5, namely, we want to find $\langle X \rangle$ and $\langle P \rangle$ by writing X and P

in terms of a and $a^\dagger$. It's easy to see from the definitions that

$$a + a^\dagger = 2 \cdot \sqrt{\frac{m\omega}{2\hbar}} X, \quad a - a^\dagger = 2 \cdot \sqrt{\frac{m\omega}{2\hbar}} \cdot \frac{i}{m\omega} P$$

so here we

$$X = \sqrt{\frac{\hbar}{2m\omega}} (a + a^\dagger), \quad P = -i \sqrt{\frac{\hbar m\omega}{2}} (a - a^\dagger).$$

To find $\langle X \rangle = \langle \psi | X | \psi \rangle$, we'll need to compute $X|\psi\rangle$ first. Since X is given in terms of a and $a^\dagger$, and $|\psi\rangle$ is terms of $|0\rangle$ and $|1\rangle$, let's do the following operators first:

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$$a|0\rangle = 0 \text{ (recall } a \text{ annihilates the ground state of the QMSHO)}$$

$$a|1\rangle = \sqrt{1}|0\rangle = |0\rangle$$

$$a^\dagger|0\rangle = \sqrt{0+1}|0+1\rangle = |1\rangle$$

$$a^\dagger|1\rangle = \sqrt{1+1}|1+1\rangle = \sqrt{2}|2\rangle$$

Thus,

$$\Sigma|\psi\rangle = \sqrt{\frac{\hbar}{2m\omega}} (a+a^\dagger) \left(\frac{\sqrt{3}}{2}|0\rangle - \frac{i}{2}|1\rangle \right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left(\frac{\sqrt{3}}{2}a|0\rangle - \frac{i}{2}a|1\rangle + \frac{\sqrt{3}}{2}a^\dagger|0\rangle - \frac{i}{2}a^\dagger|1\rangle \right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left(\frac{\sqrt{3}}{2} \cdot 0 - \frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle - \frac{i}{2}\sqrt{2}|2\rangle \right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left(-\frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle - \frac{i}{\sqrt{2}}|2\rangle \right)$$

Now, $\langle\psi| = |\psi\rangle^\dagger$, or

$$\langle\psi| = \left(\frac{\sqrt{3}}{2}|0\rangle - \frac{i}{2}|1\rangle \right)^\dagger = \frac{\sqrt{3}}{2}\langle 0| + \frac{i}{2}\langle 1|$$

also

$$\langle X \rangle = \langle\psi|\Sigma|\psi\rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left(\frac{\sqrt{3}}{2}\langle 0| + \frac{i}{2}\langle 1| \right) \left(-\frac{i}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle - \frac{i}{\sqrt{2}}|2\rangle \right)$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \left(-\frac{i\sqrt{3}}{4}\langle 0|0\rangle + \frac{1}{4}\langle 1|0\rangle + \frac{3}{4}\langle 0|1\rangle + \frac{i\sqrt{3}}{4}\langle 1|1\rangle - \frac{i\sqrt{3}}{2\sqrt{2}}\langle 0|2\rangle + \frac{1}{2\sqrt{2}}\langle 1|2\rangle \right)$$

But $\langle n|n'\rangle = 0$ if $n \neq n'$ and 1 if $n=n'$, so all but the first and fourth terms are zero, so

$$\langle X \rangle = \sqrt{\frac{\hbar}{2m\omega}} \left(-\frac{i\sqrt{3}}{4} \cdot 1 + \frac{i\sqrt{3}}{4} \cdot 1 \right) = \boxed{0}$$

[15pts]

For $\langle P \rangle$, we do a very similar calculation:

$$P|\psi\rangle = -i\sqrt{\frac{\hbar m\omega}{2}} (a-a^\dagger) \left(\frac{\sqrt{3}}{2}|0\rangle - \frac{i}{2}|1\rangle \right)$$

$$= -i\sqrt{\frac{\hbar m\omega}{2}} \left(\frac{\sqrt{3}}{2}a|0\rangle - \frac{i}{2}a|1\rangle - \frac{\sqrt{3}}{2}a^\dagger|0\rangle + \frac{i}{2}a^\dagger|1\rangle \right)$$

$$= -i\sqrt{\frac{\hbar m\omega}{2}} \left(\frac{\sqrt{3}}{2} \cdot 0 - \frac{i}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle + \frac{i}{2}\sqrt{2}|2\rangle \right)$$

$$= -i\sqrt{\frac{m\hbar\omega}{2}} \left(-\frac{i}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle + \frac{i}{\sqrt{2}}|2\rangle \right) \quad (6)$$

and thus

$$\langle p \rangle = \langle \psi | p | \psi \rangle$$

$$= \left(\frac{\sqrt{3}}{2}\langle 0| + \frac{i}{2}\langle 1| \right) \left(-i\sqrt{\frac{m\hbar\omega}{2}} \left(-\frac{i}{2}|0\rangle - \frac{\sqrt{3}}{2}|1\rangle + \frac{i}{\sqrt{2}}|2\rangle \right) \right)$$

$$= -i\sqrt{\frac{m\hbar\omega}{2}} \left(-\frac{\sqrt{3}}{4}\langle 0|0\rangle + \frac{1}{4}\langle 1|0\rangle - \frac{3}{4}\langle 0|1\rangle - \frac{i\sqrt{3}}{4}\langle 1|1\rangle \right. \\ \left. + \frac{i\sqrt{3}}{2\sqrt{2}}\langle 0|2\rangle - \frac{1}{2\sqrt{2}}\langle 1|2\rangle \right)$$

$$= -i\sqrt{\frac{m\hbar\omega}{2}} \left(-\frac{i\sqrt{3}}{4} \cdot 1 - \frac{i\sqrt{3}}{4} \cdot 1 \right)$$

$$= -i\sqrt{\frac{m\hbar\omega}{2}} \left(-\frac{i\sqrt{3}}{4} \right) = \boxed{-\sqrt{\frac{3m\hbar\omega}{32}}}$$

(15pts)

so while the average position of this state is in the middle, it's got an average momentum to the left of size $\sqrt{\frac{3m\hbar\omega}{32}}$.